

Optimization with ordering

Ordered Operators and Decision Making (I)

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Aggregation Operators

Basic Properties of Ordered Operators

Linear Ordered Measures

Quadratic Ordered Measures

Nested Ordered Measures

Computational Insights

Aggregation Operators

[illegible]

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Aggregation in optimization

Most optimization problems do not produce a vector:

$$\mathbf{x} = (x_1, \dots, x_n)$$

where the components may represent:

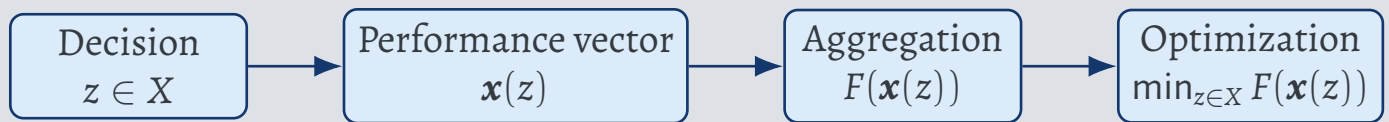
- costs under scenarios,
- distances to users,
- losses of observations,
- delays in a network,
- risks of assets,
- errors of predictions.

But Optimization needs

- one scalar objective (unless MO),
- interpretable trade-offs,
- robustness,
- fairness,
- balance,
- risk control.

The choice of aggregation changes the meaning of the optimization model.

From a feasible decision to many outcomes



The aggregation is not
a post-processing step:
it is embedded in the decision process.

Classical aggregation paradigms

Average behavior

$$\frac{1}{n} \sum_{i=1}^n x_i$$

Efficient on average.

Worst case

$$\max_{i=1,\dots,n} x_i$$

Protects against extremes.

Best case

$$\min_{i=1,\dots,n} x_i$$

Focuses on the most favorable outcome.

These three operators are useful, but they only capture a small part of the possible distributional behavior.

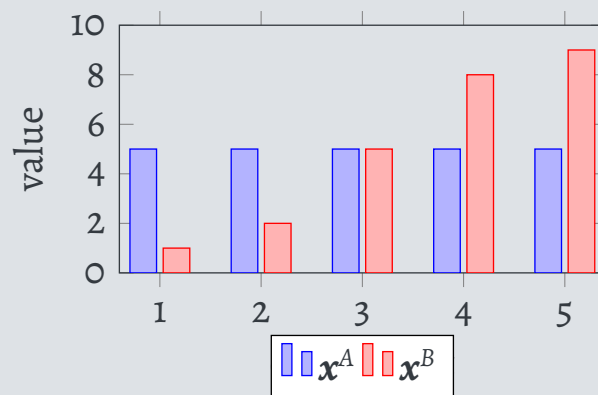
What classical aggregation may miss

Two vectors with the same mean

$$\mathbf{x}^A = (5, 5, 5, 5, 5)$$

$$\mathbf{x}^B = (1, 2, 5, 8, 9)$$

Both have mean 5.

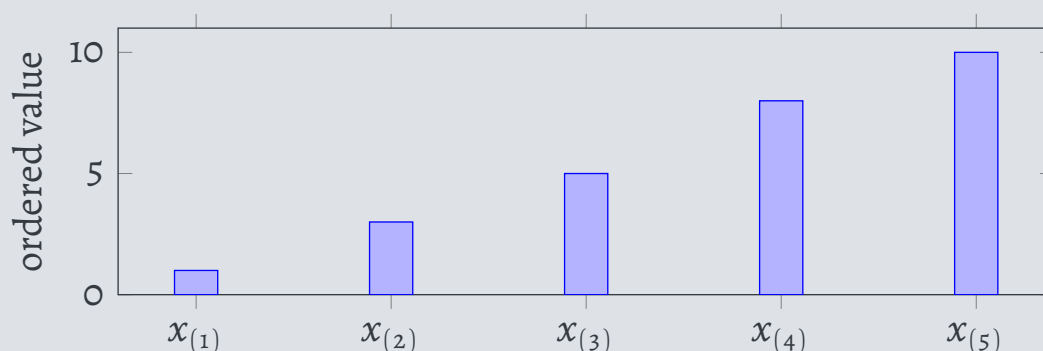


Mean aggregation ignores dispersion, imbalance, and tail behavior.

The key idea: order the values

Given a vector $\mathbf{x} = (x_1, \dots, x_n)$, define its ordered version:

$$x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n)}.$$

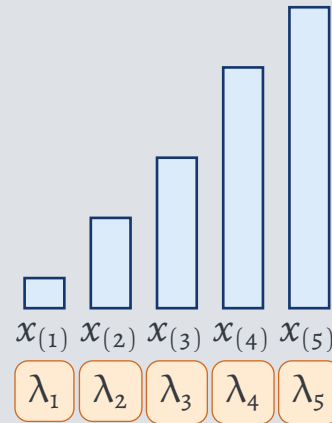


Once values are ordered, we can decide how much importance to assign to each rank.

Ordered measures as weighted sums of ranks

A linear ordered measure, or L -measure, has the form:

$$\ell(\mathbf{x}, \boldsymbol{\lambda}) = \sum_{k=1}^n \lambda_k x_{(k)}.$$



Statistic	Expression	$\boldsymbol{\lambda}$
Arithmetic mean	$\frac{1}{n} \sum_{i=1}^n x_i$	$\frac{1}{n} (1, \dots, 1)$
Weighted mean	$\frac{1}{\sum_{i=1}^n w_i} \sum_{i=1}^n w_i x_i, \quad w_i \geq 0$	$(\frac{w_1}{\sum_{i=1}^n w_i}, \dots, \frac{w_n}{\sum_{i=1}^n w_i})$
Maximum	$\max x_i$	$(0, \dots, 0, 1)$
Minimum	$\min x_i$	$(1, 0, \dots, 0, 0)$
Median	$\text{median}(x) = \begin{cases} x_{(m+1)}, & n = 2m+1 \\ \frac{x_{(m)} + x_{(m+1)}}{2}, & n = 2m \end{cases}$	$(0, \dots, 0, \underbrace{\frac{1}{2}}_{m\text{-th}}, \underbrace{\frac{1}{2}}_{(m+1)\text{-th}}, 0, \dots, 0)$
q -th quantile (or τ -quantile)	$x_{(q)}, q = \lfloor \tau n \rfloor$	$(0, \dots, 0, \underbrace{1}_{q\text{-th}}, 0, \dots, 0)$
Midrange	$\frac{1}{2} (\max x_i + \min x_i)$	$\frac{1}{2} (1, 0, \dots, 0, 1)$
General midrange (midhinge, etc.)	$\frac{1}{2} (x_{(r)} + x_{(s)}), (r \leq s)$	$\frac{1}{2} (0, \dots, 0, \underbrace{1}_{r\text{-th}}, 0, \dots, 0, \underbrace{1}_{s\text{-th}}, \dots, 0)$
(r, s) -trimmed mean	$\frac{1}{n-r-s} \sum_{i=r+1}^{n-s} x_{(i)}$	$\frac{1}{n-r-s} (0, \dots, 0, \underbrace{1}_r, \dots, \underbrace{1}_{n-r-s}, \underbrace{0}_s, \dots, 0)$
(r, s) -mid mean	$\frac{1}{r} \sum_{i=1}^r x_{(i)} + \frac{1}{s} \sum_{i=n-s}^n x_{(i)}$	$(\underbrace{\frac{1}{r}, \dots, \frac{1}{r}}_r, \underbrace{0, \dots, 0}_{n-r-s}, \underbrace{\frac{1}{s}, \dots, \frac{1}{s}}_s)$
(r, s) -winsorized mean	$\frac{1}{n} (rx_{(r+1)} + \sum_{i=r+1}^{n-s} x_{(i)} + sx_{(n-s)})$	$\frac{1}{n} (0, \dots, 0, \underbrace{r+1}_{(r+1)\text{-th}}, 1, \dots, 1, \underbrace{s+1}_{(n-s)\text{-th}}, 0, \dots, 0)$

The vector $\boldsymbol{\lambda}$ is a modeling language.

Particular cases... widely studied

	Operator	References
Extremal	Minimax, centdian, k -centrum, k -sum optimization	Schrijver (1982); Hansen and Labbé (1988); Garfinkel et al. (2006); Tamir (2001); Puerto et al. (2017); Li et al. (2022); Bucarey et al. (2025)
Equity/fairness	Minimum envy solutions, inequality and equity indices	Espejo et al. (2009); Blanco et al. (2024); Marsh and Schilling (1994); Sen and Foster (1997); Young (1995); Bertsimas et al. (2011)
Dispersion	Minimum deviation, variance-type and range-based criteria	Gupta and Punnen (1988); Maimon (1986); Punnen and Aneja (1997); Dimnaku et al. (2005)
Risk-oriented	Regret models, VaR and CVaR	Averbakh (2001); Mansini et al. (2007); Scutella and Recchia (2013); Daskin et al. (2024)

Dispersion Measures and Equity

- **Dispersion measures** quantify how spread or heterogeneous a set of values is.
- They typically:
 - attain value 0 when all observations coincide,
 - increase as the variability of the data increases.
- Classical examples: Variance, Standard deviation, Mean absolute deviation.
- Most of these measures are computed via deviations from a *reference* tendency measure:

$$\text{Dispersion} \approx \sum_i \text{Deviation}(x_i, \text{center})$$

- Robust alternatives reduce the influence of outliers: Interquartile range, Median absolute deviation.
- Dispersion measures are also linked to **fairness and equity**: deviations from a reference point (Hoover, 1941), pairwise inequality comparisons (Gini, 1921).

Statistic	Expression	λ
Range	$\max x_i - \min x_i$	$(-1, 0, \dots, 0, 1)$
General interquantile range (e.g., interquartile or interdecile ranges)	$x_{(r)} - x_{(s)}, (r < s)$	$(0, \dots, 0, \underbrace{-1}_{r\text{-th}}, 0, \dots, 0, \underbrace{1}_{s\text{-th}}, \dots, 0)$
Variance (as squared deviation from the mean)	$\frac{1}{n} \sum_{i=1}^n (x_i - \frac{1}{n} \sum_{j=1}^n x_j)^2$	
Mean squared deviation from a location measure $m(x)$ (e.g., from the q -th quantile)	$\frac{1}{n} \sum_{i=1}^n (x_i - m(x))^2$	
Gini differences or mean absolute difference	$\frac{1}{n^2} \sum_{i,j=1}^n x_i - x_j $	$\frac{2}{n^2} ((2i - n - 1))_{i \in N}$
Mean absolute deviation from mean	$\frac{1}{n} \sum_{i=1}^n x_i - \frac{1}{n} \sum_{j=1}^n x_j $	
Mean absolute deviation from a location measure $m(x)$ (e.g., from the q -th quantile)	$\frac{1}{n} \sum_{i=1}^n x_i - m(x) $	$\frac{1}{n} (-1, \dots, -1, \underbrace{2q - n - 1}_{q\text{-th}}, 1, \dots, 1)$ (e.g., $m(x) = x_{(q)}$)

OWA Operators and Ordered Aggregation

- Ordered aggregation operators arise naturally in:
 - Statistics,
 - Decision Theory,
 - Data Science,
 - Operations Research.
- Classical **OWA (Ordered Weighted Average)** operators were introduced by **Yager (1988)**.
- Standard OWA operators aggregate ordered values through:

$$\sum_{i=1}^n \lambda_i x_{(i)}, \quad \lambda_i \geq 0, \quad \sum_i \lambda_i = 1.$$

Applications of OWA Operators in Data Science and OR

Application Area	Representative References
Fairness-aware optimization	(Mesa et al., 2003; Ogryczak, 2009; Blanco et al., 2014; Tsang and Shehadeh, 2025)
Robust regression and estimators	(Yager and Beliakov, 2009; Blanco et al., 2018; Flores-Sosa et al., 2020; Blanco et al., 2021; Flores-Sosa et al., 2022; D'Urso and Chachi, 2022; Puerto and Torrejon, 2025)
Support Vector Machines	(Maldonado et al., 2018; Marín et al., 2022)
Bayesian insurance premiums	(Blanco and Pérez-Sánchez, 2018)
Neural nets and fuzzy quantifiers	(Yager, 1987, 1992; Li et al., 2021)
Image segmentation	(Calvino et al., 2022; Muñoz-Ocaña et al., 2024)
Clustering	(Cheng et al., 2009)
Facility and hub location	(Nickel and Puerto, 2006; Puerto and Rodríguez-Chía, 2019; Labbé et al., 2017; Marín et al., 2020; Blanco and Gázquez, 2023; Ljubić et al., 2024)
Voting and social choice	(Amanatidis et al., 2015; Ponce et al., 2018)
Portfolio selection and network design	(Cesarone and Puerto, 2024; Puerto et al., 2016; Pozo et al., 2021, 2024)

Ordered Operators

- Despite their different origins and applications, all these operators share a **common ordered structure**.
- Many of them are naturally formulated through **linear, convex, or mixed-integer optimization models**.
- The classification is not unique: some measures may admit multiple interpretations (e.g., inequality measures as dispersion criteria).

Fairness and equity

In many problems, the components of $\mathbf{x}$ represent agents, users, regions, customers, or scenarios.

Efficiency objective

$$\min \sum_i x_i$$

May produce very uneven outcomes.

Ordered fairness objective

$$\min \sum_k \lambda_k x_{(k)}$$

Can emphasize the worst-off or most exposed components.

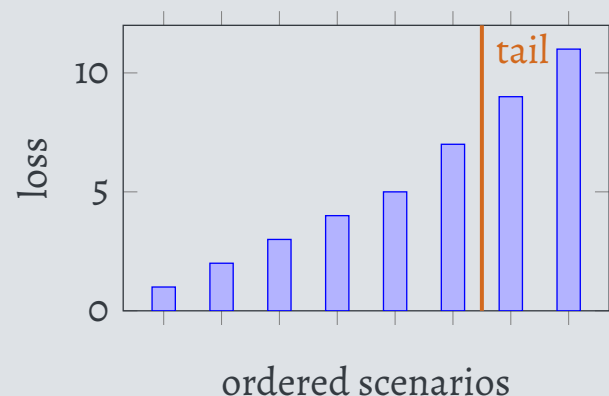


Risk-sensitive decision making

Ordered measures naturally describe tails of distributions.

Typical ordered risk criteria

- maximum loss,
- quantiles,
- Value-at-Risk type measures,
- Conditional Value-at-Risk type measures,
- regret and tail aggregation.



Ordering makes tail behavior explicit.

Orness and Andness (Yager, 1988)

$$\text{orness}(\lambda) = \frac{1}{n-1} \sum_{i=1}^n (n-i)\lambda_i \quad \text{andness}(\lambda) = 1 - \text{orness}(\lambda).$$

High Orness (≈ 1): Weights concentrated on large ordered values ($\lambda = (0, \dots, 0, 1)$).

- Optimistic behavior.
- Rewards outstanding performance.
- Strong compensability.

High Andness (≈ 1): Weights concentrated on small ordered values ($\lambda = (1, 0, \dots, 0)$).

- Pessimistic behavior.
- Focuses on worst performance.
- Low compensability.

Operator	Minimum	Average	Maximum
Orness	0	0.5	1

Note: The orness/andness indices quantify the decision maker's attitude toward compensation between criteria, interpolating continuously between conjunctive (AND-like) and disjunctive (OR-like) aggregation.

Basic Properties of Ordered Operators

Basic Analytical Properties

$$\ell(\mathbf{x}, \lambda) = \sum_{k=1}^n \lambda_k x_{(k)}, \quad x_{(1)} \leq \dots \leq x_{(n)}.$$

Algebraic properties

- Symmetric: $\ell(\mathbf{x}, \lambda) = \ell(\pi(\mathbf{x}), \lambda)$, for every permutation π .
- Positively homogeneous:
 $\ell(\alpha \mathbf{x}, \lambda) = \alpha \ell(\mathbf{x}, \lambda)$, $\alpha \geq 0$.
- Translation invariant:
 $\ell(\mathbf{x} + \beta \mathbf{1}, \lambda) = \ell(\mathbf{x}, \lambda) + \beta \sum_{k=1}^n \lambda_k$.

Order properties

- If $\mathbf{x} \leq \mathbf{y}$, and $\lambda_k \geq 0$, then
 $\ell(\mathbf{x}, \lambda) \leq \ell(\mathbf{y}, \lambda)$.
- If additionally $\sum_{k=1}^n \lambda_k = 1$, then:
 $x_{(1)} \leq \ell(\mathbf{x}, \lambda) \leq x_{(n)}$.
- $\ell(\mathbf{x}, \lambda) \in \text{conv}\{x_1, \dots, x_n\}$.

Majorization and Schur Convexity

Majorization

$\mathbf{x} \succeq \mathbf{y}$ iff:

- $\sum_{i=k}^n x_{(i)} \geq \sum_{i=k}^n y_{(i)}$, for $k = 2, \dots, n$,
and
 - $\sum_{i=1}^n x_i = \sum_{i=1}^n y_i$.
- $\mathbf{x} \succeq \mathbf{y}$: $\mathbf{x}$ is more dispersed than $\mathbf{y}$.

Schur convexity: A function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is Schur-convex if

$$\mathbf{x} \succeq \mathbf{y} \Rightarrow f(\mathbf{x}) \geq f(\mathbf{y}).$$

$$\mathbf{x} \succeq \mathbf{y} \Rightarrow f(\mathbf{x}) \leq f(\mathbf{y})$$

defines Schur-concavity.

Theorem

- $\lambda_1 \leq \dots \leq \lambda_n \Rightarrow \ell$ is Schur-convex.
- $\lambda_1 \geq \dots \geq \lambda_n \Rightarrow \ell$ is Schur-concave.
- If $\lambda_1 \leq \dots \leq \lambda_n$, then ℓ rewards dispersion: ℓ assigns greater value to solutions with more extreme large components.
- If $\lambda_1 \geq \dots \geq \lambda_n$, then ℓ penalizes dispersion: ℓ favors balanced and equitable distributions.

<https://arxiv.org/abs/2503.18097>

Linear Ordered Measures

$$\ell(\mathbf{x}, \boldsymbol{\lambda}) = \sum_{k=1}^n \lambda_k x_{(k)},$$

where $x_{(1)}, \dots, x_{(n)}$ denote the components of $\mathbf{x}$ ordered in non increasing order.

- The standard approach is to sort $\mathbf{x}$ and evaluate the weighted sum:

Algorithm	Best	Average	Worst
Bubble Sort	$O(n)$	$O(n^2)$	$O(n^2)$
Selection Sort	$O(n^2)$	$O(n^2)$	$O(n^2)$
Insertion Sort	$O(n)$	$O(n^2)$	$O(n^2)$
Merge Sort	$O(n \log n)$	$O(n \log n)$	$O(n \log n)$
Quick Sort	$O(n \log n)$	$O(n \log n)$	$O(n^2)$
Heap Sort	$O(n \log n)$	$O(n \log n)$	$O(n \log n)$
TimSort	$O(n)$	$O(n \log n)$	$O(n \log n)$

$\Rightarrow O(n \log n) + O(n)$

- But the same value can be computed through linear optimization models
- This will allow ordered measures to be embedded inside opt problems.
- In some cases, the resulting formulations can be computationally competitive with sorting-based implementations.

Mean, Maximum and Minimum as LPs

Mean

$$\text{mean}(\mathbf{x}) = \min \gamma$$

$$\begin{aligned} \text{s.t. } & \sum_{i \in N} x_i \leq n\gamma, \\ & \gamma \in \mathbb{R}. \end{aligned}$$

Maximum

$$\max(\mathbf{x}) = \min \gamma$$

$$\begin{aligned} \text{s.t. } x_i &\leq \gamma, \quad i \in N, \\ \gamma &\in \mathbb{R}. \end{aligned}$$

Minimum

$$\min(\mathbf{x}) = \max \gamma$$

$$\begin{aligned} \text{s.t. } x_i &\geq \gamma, \quad i \in N, \\ \gamma &\in \mathbb{R}. \end{aligned}$$

- In all three cases, the constraints impose bounds on γ .
- The objective direction forces the corresponding bound to be tight.
- Hence, the optimal value of γ is exactly the aggregation operator.

But... General Ordered Measures Are More Difficult

For the mean, maximum and minimum, the aggregation can be described by a single auxiliary variable γ .

For a general linear ordered measure:

$$\ell(\mathbf{x}, \lambda) = \sum_{k=1}^n \lambda_k x_{(k)},$$

- the value attached to position k is not known in advance;
- one must decide which original component x_i occupies each ordered position;
- the ordering constraints must ensure that the selected values form a sorted sequence

Thus, computing $\ell(\mathbf{x}, \lambda)$ by optimization requires modeling the sorting operation.

Assignment-based Integer Formulation (Boland et al., 2006)

$$z_{ik} = \begin{cases} 1, & \text{if component } i \text{ is assigned to ordered position } k, \\ 0, & \text{otherwise.} \end{cases}$$

$$\begin{aligned} \ell_1(\mathbf{x}, \lambda) = \min & \sum_{i=1}^n \sum_{k=1}^n \lambda_k z_{ik} x_i \\ \text{s.t.} & \sum_{i=1}^n z_{ik} = 1, & k \in N, \\ & \sum_{k=1}^n z_{ik} = 1, & i \in N, \\ & \sum_{i=1}^n z_{ik} x_i \leq \sum_{i=1}^n z_{i,k+1} x_i, & k \in N, k < n, \\ & z_{ik} \in \{0, 1\}, & i, k \in N. \end{aligned}$$

Assignment-based Integer Formulation

- $\sum_{i=1}^n z_{ik} = 1$: each ordered position receives exactly one component.
- $\sum_{k=1}^n z_{ik} = 1$: each component is assigned to exactly one position.
- $\sum_{i=1}^n z_{ik} x_i \leq \sum_{i=1}^n z_{i,k+1} x_i$: the assigned values follow the required order.
- Once the assignment is fixed:

$$x_{(k)} = \sum_{i \in N} z_{ik} x_i.$$

- The objective function becomes: $\sum_{k=1}^n \lambda_k x_{(k)}$.

Limitation. The model has $\mathcal{O}(n^2)$ binary variables, which may be difficult for large datasets.

k -sum Representation: Ordered Operators

$$x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n)}.$$

Define the k -sum operator as the sum of the largest $n - k + 1$ components:

$$S_k(\mathbf{x}) = \sum_{\ell=k}^n x_{(\ell)}, \quad k = 1, \dots, n,$$

We also define the increments

$$\lambda_0 := 0, \quad \Delta_k := \lambda_k - \lambda_{k-1}, \quad k = 1, \dots, n. \Rightarrow \lambda_k = \sum_{j=1}^k \Delta_j.$$

k -sum Representation: Telescopic Expansion

$$\begin{aligned}
 \sum_{k=1}^n \Delta_k S_k(\mathbf{x}) &= \sum_{k=1}^n (\lambda_k - \lambda_{k-1}) \left(\sum_{\ell=k}^n x_{(\ell)} \right) \\
 &= \sum_{k=1}^n (\lambda_k - \lambda_{k-1}) (x_{(k)} + x_{(k+1)} + \cdots + x_{(n)}) \\
 &= \lambda_1 (x_{(1)} + \cdots + x_{(n)}) + (\lambda_2 - \lambda_1) (x_{(2)} + \cdots + x_{(n)}) \\
 &\quad + \cdots + (\lambda_n - \lambda_{n-1}) x_{(n)} \\
 &= \lambda_1 x_{(1)} + \lambda_2 x_{(2)} + \cdots + \lambda_n x_{(n)} \\
 &= \sum_{k=1}^n \lambda_k x_{(k)} = \ell(\mathbf{x}, \boldsymbol{\lambda}).
 \end{aligned}$$

Computing the k -sum Operator: Selection Model

$$S_k(\mathbf{x}) = \sum_{\ell=k}^n x_{(\ell)}, \quad k = 1, \dots, n,$$

Consider the binary variables

$$d_i = \begin{cases} 1, & \text{if component } x_i \text{ is selected in the sum,} \\ 0, & \text{otherwise.} \end{cases}$$

Since we want to select exactly $n - k + 1$ components, we impose

$$\sum_{i=1}^n d_i = n - k + 1.$$

Computing the k -sum Operator: Primal Formulation

Selecting the largest $n - k + 1$ components can be written as

$$\begin{aligned}
 S_k(\mathbf{x}) &= \max \sum_{i=1}^n d_i x_i \\
 \text{s.t. } &\sum_{i=1}^n d_i = n - k + 1, \\
 &d_i \in \{0, 1\} [0, 1], \quad \forall i, j \in [n].
 \end{aligned}$$

$$\text{Extremes} \left(\left\{ \mathbf{d} \in [0, 1]^n : \sum_{i=1}^n d_i = n - k + 1 \right\} \right) \in \{0, 1\}^n$$

Computing the k -sum Operator: Dual Formulation

The primal problem is

$$\begin{aligned}
 S_k(\mathbf{x}) &= \max \sum_{i=1}^n d_i x_i \\
 \text{s.t. } &\sum_{i=1}^n d_i = n - k + 1, \\
 &d_i \leq 1, \quad i \in N, \\
 &d_i \geq 0, \quad i \in N.
 \end{aligned}$$

The dual problem is

$$\begin{aligned}
 S_k(\mathbf{x}) &= \min (n - k + 1)t + \sum_{i=1}^n v_i \\
 \text{s.t. } &t + v_i \geq x_i, \quad i \in N, \\
 &t \in \mathbb{R}, \\
 &v_i \geq 0, \quad i \in N.
 \end{aligned}$$

Computing the k -sum Operator

By Strong duality:

Primal formulation

$$S_k(\mathbf{x}) = \max \sum_{i=1}^n d_i x_i$$

$$\text{s.t. } \sum_{i=1}^n d_i = n - k + 1,$$

$$0 \leq d_i \leq 1, \quad i \in N.$$

Dual formulation

$$S_k(\mathbf{x}) = \min (n - k + 1)t + \sum_{i=1}^n v_i$$

$$\text{s.t. } t + v_i \geq x_i, \quad i \in N,$$

$$t \in \mathbb{R},$$

$$v_i \geq 0, \quad i \in N.$$

- The primal selects the $n - k + 1$ largest components.
- The dual represents the same value without explicit selection.
- This representation is the basis of a continuous linear model.

Monotone λ

Recall that:

$$\ell(\mathbf{x}, \lambda) = \sum_{k=1}^n \Delta_k(\mathbf{x})$$

If all $\Delta_k \geq 0$ (equivalently $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$)

$$\ell(\mathbf{x}, \lambda) = \sum_{k=1}^n \Delta_k S_k(\mathbf{x}) = \min \sum_{k=1}^n \Delta_k \left((n - k + 1)t_k + \sum_{i=1}^n v_{ik} \right)$$

$$\text{s.t. } t_k + v_{ik} \geq x_i, \quad i \in N,$$

$$t_k \in \mathbb{R}, \quad k \in N,$$

$$v_{ik} \geq 0, \quad i, k \in N.$$

Convexity

Theorem

$\ell(\mathbf{x}, \lambda)$ is convex (in $\mathbf{x}$) if and only if $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$, and concave if and only if $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$.

Proof sketch. Using that $\ell(\mathbf{x}, \lambda) = \sum_{k=1}^n \Delta_k S_k(\mathbf{x})$ since each S_k is convex, ℓ is convex whenever

$$\Delta_r \geq 0 \iff \lambda_1 \leq \dots \leq \lambda_n.$$

Conversely, if $\lambda_s > \lambda_{s+1}$ for some s , suitable vectors $\mathbf{c}, \mathbf{d} \in \mathbb{R}^n$ satisfy

$$\ell(\mathbf{c} + \mathbf{d}, \lambda) > \ell(\mathbf{c}, \lambda) + \ell(\mathbf{d}, \lambda),$$

so ℓ is not subadditive. Since ℓ is positively homogeneous, convexity fails ([Rockafellar, 1970](#)).

Convexity

Convex case

- Increasing weights emphasize large order statistics.
- Includes:
 - maximum,
 - top- k sums,
 - CVaR-type operators.
- Suitable for:
 - robustness,
 - risk aversion,
 - worst-case optimization.

Concave case

- Decreasing weights emphasize small order statistics.
- Includes:
 - minimum,
 - lower trimmed means.
- Suitable for:
 - equitable optimization,
 - balanced allocations,
 - fairness criteria.

Break: $\lambda_1 \leq \lambda_2 \leq \dots \leq \lambda_n$

$$\ell(\mathbf{x}, \lambda) = \sum_{k=1}^n \lambda_k x_{(k)} = \max_{\sigma \in \mathcal{P}_n} \sum_{k=1}^n \lambda_k x_{\sigma(k)}$$

$\mathcal{P}_n$: Permutations of $\{1, \dots, n\}$.

$$\ell(\mathbf{x}, \lambda) = \max \sum_{k=1}^n \sum_{i=1}^n \lambda_k x_i z_{ik}$$

$$\text{s.t.} \quad \sum_{k=1}^n z_{ik} = 1, \forall i = 1, \dots, n,$$

$$\sum_{i=1}^n z_{ik} = 1, \forall k = 1, \dots, n,$$

$$z_{ik} \in \{0, 1\} [0, 1], \forall i, k = 1, \dots, n.$$

BEP (Blanco et al., 2014)

$$\begin{aligned} \min \quad & \sum_{i=1}^n u_i + \sum_{k=1}^n v_k \\ \text{s.t.} \quad & u_i + v_k \geq \lambda_k x_i, \quad \forall i, k = 1, \dots, n, \\ & u_i \in \mathbb{R}, \quad \forall i = 1, \dots, n, \\ & v_k \in \mathbb{R}, \quad \forall k = 1, \dots, n. \end{aligned}$$

General case again...

$\Delta_k > 0$:

$$\begin{aligned} \Delta_k S_k(\mathbf{x}) &= \max \Delta_k \sum_{i=1}^n d_{ik} x_i \\ \text{s.t.} \quad & \sum_{i=1}^n d_{ik} = n - k + 1, \\ & 0 \leq d_{ik} \leq 1, \quad i \in N. \end{aligned}$$

$$\begin{aligned} \min \Delta_k \quad & \left((n - k + 1) t_k + \sum_{i=1}^n v_{ik} \right) \\ \text{s.t.} \quad & t_k + v_{ik} \geq x_i, \quad i \in N, \\ & t_k \in \mathbb{R}, \\ & v_{ik} \geq 0, \quad i \in N. \end{aligned}$$

$\Delta_k < 0$:

$$\begin{aligned} \Delta_k S_k(\mathbf{x}) &= \Delta_k \max \sum_{i=1}^n d_{ik} x_i \\ \text{s.t.} \quad & \sum_{i=1}^n d_{ik} = n - k + 1, \\ & 0 \leq d_{ik} \leq 1, \quad i \in N. \end{aligned}$$

$$\begin{aligned} \min \Delta_k \quad & \sum_{i=1}^n d_{ik} x_i \\ \text{s.t.} \quad & \sum_{i=1}^n d_{ik} = n - k + 1, \\ & 0 \leq d_{ik} \leq 1, \quad i \in N. \end{aligned}$$

k -sum for General Case

$$\Delta^+ = \{j \in N : \Delta_j > 0\}, \quad \Delta^- = \{j \in N : \Delta_j < 0\}$$

$$\begin{aligned} \ell_2(\mathbf{x}, \lambda) = \min & \sum_{k \in \Delta^+} \Delta_k \left((n - k + 1)t_k + \sum_{i=1}^n v_{ik} \right) + \sum_{k \in \Delta^-} \Delta_k \sum_{i=1}^n d_{ik} x_i \\ \text{s.t. } & t_k + v_{ik} \geq x_i, & i \in N, k \in \Delta^+, \\ & \sum_{i=1}^n d_{ik} = n - k + 1, & k \in \Delta^-, \\ & t_k \in \mathbb{R}, \quad v_{ik} \geq 0, & i \in N, k \in \Delta^+, \\ & 0 \leq d_{ik} \leq 1, & i \in N, k \in \Delta^-. \end{aligned}$$

Assignment-based vs. k -sum Formulation

The model $\ell_2(\mathbf{x}, \lambda)$ provides an important improvement over the assignment formulation.

- It avoids binary variables.
- It is a linear optimization model.
- It has $\mathcal{O}(n^2)$ continuous variables and $\mathcal{O}(n^2)$ constraints in the worst case.
- It can be decomposed into independent parts associated with Δ^+ and Δ^- .
- When only a few increments Δ_k are nonzero, the model can be much smaller.

Note: One can replace the *convex* part by **BEP**...

Quantile-based Representation

For $0 < \tau < 1$, define the sample τ -quantile as

$$\text{quantile}_{\tau}(\mathbf{x}) = \arg \min_{\gamma \in \mathbb{R}} \mathcal{L}_{\tau}(\mathbf{x}, \gamma),$$

where

$$\mathcal{L}_{\tau}(\mathbf{x}, \gamma) = \tau \sum_{i: x_i \geq \gamma} (x_i - \gamma) + (1 - \tau) \sum_{i: x_i < \gamma} (\gamma - x_i).$$

- If $x_i > \gamma$, the deviation is penalized with weight τ .
- If $x_i < \gamma$, the deviation is penalized with weight $1 - \tau$.

Quantile-based Representation

For $0 < \tau < 1$, define the sample τ -quantile as

$$\text{quantile}_{\tau}(\mathbf{x}) = \arg \min_{\gamma \in \mathbb{R}} \mathcal{L}_{\tau}(\mathbf{x}, \gamma),$$

where

$$\mathcal{L}_{\tau}(\mathbf{x}, \gamma) = \tau \sum_{i: x_i \geq \gamma} (x_i - \gamma) + (1 - \tau) \sum_{i: x_i < \gamma} (\gamma - x_i).$$

When minimizing $\mathcal{L}$ on γ : $\#\{i : x_i < \gamma\} \approx \tau n$, $\#\{i : x_i > \gamma\} \approx (1 - \tau)n$.

- $\tau = 0.5$: symmetric absolute loss $\Rightarrow$ median.
- $\tau = 0.9$: upward deviations are penalized more $\Rightarrow$ upper-tail quantile.
- $\tau = 0.1$: downward quantile $\Rightarrow$ lower-tail quantile.

- The primal uses positive and negative deviations from the threshold γ .
- The dual gives a compact weighted representation in the original data.
- Both formulations are used to build the quantile-based ordered-measure model.

Quantile-based Formulation

$$\begin{aligned} \ell_3(\mathbf{x}, \boldsymbol{\lambda}) = & \min \sum_{k \in \Delta^+} \sum_{i \in N} \Delta_k \left(\frac{k}{n} u_{ik} + \left(1 - \frac{k}{n} \right) v_{ik} \right) + \sum_{k \in \Delta^-} \sum_{i \in N} \Delta_k \alpha_{ik} x_i \\ & + \sum_{k \in N} \sum_{i \in N} \Delta_k \left(1 - \frac{k}{n} \right) x_i \end{aligned}$$

$$\text{s.t. } u_{ik} - v_{ik} + \gamma_k = x_i,$$

$$i \in N, k \in \Delta^+,$$

$$\sum_{i \in N} \alpha_{ik} = 0,$$

$k \in \Delta^{-},$

$$\gamma_k \in \mathbb{R}, \quad u_{ik}, v_{ik} \geq 0,$$

$$i \in N, k \in \Delta^+,$$

$$-\frac{k}{n} \leq \alpha_{ik} \leq 1 - \frac{k}{n},$$

$$i \in N, k \in \Delta^-.$$

Interpretation of the Quantile-based Model

The quantile-based formulation follows the same sign decomposition principle as the k -sum model.

- For terms with $\Delta_k > 0$, the primal quantile representation is used.
- For terms with $\Delta_k < 0$, the dual quantile representation is used.
- The connection between quantile loss and k -sums allows the reconstruction of

$$\sum_{k=1}^n \Delta_k S_k(\mathbf{x}).$$

- The model is linear and contains no binary variables.
- The final constant term appears because the k -sum is obtained from the quantile loss after adding

$$(1 - \tau) \sum_{i=1}^n x_i.$$

Key idea.

Decomposition of Continuous Formulations

$$\ell_i(\mathbf{x}, \lambda) = \ell_i^+(\mathbf{x}, \lambda) - \ell_i^-(\mathbf{x}, \lambda) + c_i, \quad i \in \{2, 3\}$$

- ℓ_i^+ is the part associated with Δ^+ .
- ℓ_i^- is the part associated with Δ^- .
- for the k -sum formulation: $c_2 = \circ$.
- for the quantile formulation: $c_3 = \sum_{k \in \Delta^+} \sum_{i \in N} \Delta_k \left(1 - \frac{k}{n}\right) x_i - \sum_{k \in \Delta^-} \sum_{i \in N} \Delta_k x_i$.

Size of the Optimization Models

Model	Binary Variables	Continuous Variables	Constraints
ℓ_1	$\mathcal{O}(n^2)$	$\circ$	$\mathcal{O}(n)$
ℓ_2	$\circ$	$\mathcal{O}(n^2)$	$\mathcal{O}(n^2)$
ℓ_3	$\circ$	$\mathcal{O}(n^2)$	$\mathcal{O}(n^2)$

- ℓ_1 is conceptually simple but uses binary assignment variables.
- ℓ_2 and ℓ_3 are continuous linear optimization models.
- The reported dimensions are worst-case bounds.
- If only a few increments Δ_k are nonzero, then $|\Delta^+|$ and $|\Delta^-|$ may be small.
- In that case, the continuous formulations may reduce to $\mathcal{O}(n)$ variables and constraints.

Quadratic Ordered Measures

V. Blanco, M.A. Pozo, J. Puerto, and A. Torrejón

<https://arxiv.org/abs/2503.18097>

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QUADRATIC ORDERED MEASURES

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Quadratic Ordered Measures

Assuming again: $x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n)}$.

$$\mathcal{Q}(\mathbf{x}, M) = \sum_{i=1}^n \sum_{j=1}^n m_{ij} x_{(i)} x_{(j)}.$$

- $M = (m_{ij})_{i,j=1}^n$ encodes interactions between ordered positions.
- The value depends on the sorted vector, not on the original labels.
- Hence, $\mathcal{Q}(\mathbf{x}, M)$ is invariant under permutations of $\mathbf{x}$.

Linear

Weight ordered components

Quadratic

Weight pairwise interactions between ordered components.

Why Quadratic Ordered Measures?

Quadratic ordered measures allow terms such as

$$\mathcal{X}_{(1)}\mathcal{X}_{(n)}$$

Local interaction

$$\sum_{i=1}^{n-1} \mathbf{x}_{(i)} \mathbf{x}_{(i+1)}$$

- They capture dispersion, scale, and variability.
- They model interactions between ordered positions.
- They include variance, trimmed variance, and winsorized variance.
- Higher-order interactions could be modeled with tensors.

Basic Properties

$$\mathcal{Q}(\mathbf{x}, M) = \mathbf{x}_{()}^\top M \mathbf{x}_{()}, \quad \mathbf{x}_{()} = (x_{(1)}, \dots, x_{(n)})^\top.$$

- $\mathcal{Q}$ is continuous in $\mathbf{x}$.
- $\mathcal{Q}$ is symmetric with respect to permutations of the original components.
- If $M \succeq 0$, then $\mathcal{Q}$ is convex in the ordered vector $\mathbf{x}_{()}$.
- If $M \succ 0$, then the quadratic form is strictly convex in $\mathbf{x}_{()}$.

Squared Deviation from an Ordered Mean

$$\ell(\mathbf{x}, \lambda) = \sum_{k=1}^n \lambda_k x_{(k)} \Rightarrow \sum_{i=1}^n (x_i - \ell(\mathbf{x}, \lambda))^2.$$

$$\sum_{i=1}^n (x_i - \ell(\mathbf{x}, \lambda))^2 = \sum_{i=1}^n \left(x_{(i)} - \sum_{j=1}^n \lambda_j x_{(j)} \right)^2.$$

It measures dispersion around an ordered linear location measure.

Squared Deviation as a $\mathcal{Q}$ -Measure

$$\mathcal{Q}(\mathbf{x}, M) = \sum_{i=1}^n \sum_{j=1}^n m_{ij} x_{(i)} x_{(j)}.$$

$$\sum_{i=1}^n \left(x_{(i)} - \sum_{j=1}^n \lambda_j x_{(j)} \right)^2 = \sum_{i=1}^n (1 + n\lambda_i^2 - 2\lambda_i) x_{(i)}^2 + \sum_{\substack{i,j=1 \\ i \neq j}}^n (n\lambda_i \lambda_j - 2\lambda_j) x_{(i)} x_{(j)}.$$

Thus, it is obtained from $\mathcal{Q}(\mathbf{x}, M)$ with

$$m_{ij} = \begin{cases} 1 + n\lambda_i^2 - 2\lambda_i, & i = j, \\ n\lambda_i \lambda_j - 2\lambda_j, & i \neq j. \end{cases}$$

- The diagonal terms measure squared ordered components.
- The off-diagonal terms encode interactions induced by the reference location.

- The lowest r and largest s observations are discarded.
- Only the central ordered block contributes to the quadratic form.

Winsorized Variance

Instead of discarding extremes, winsorization replaces them by boundary values.

(r, s) -winsorized sample

$$\underbrace{x_{(r+1)}, \dots, x_{(r+1)}}_{r+1 \text{ times}}, \quad x_{(r+2)}, \dots, x_{(n-s-1)}, \quad \underbrace{x_{(n-s)}, \dots, x_{(n-s)}}_{s+1 \text{ times}}.$$

The corresponding variance is also a $\mathcal{Q}$ -measure.

- Boundary ordered values receive larger diagonal weights.
- Interactions involving the boundary values are modified.
- Positions outside the winsorized range receive zero weight.

Optimization View: Sorting by Assignment

As in the linear case, introduce

$$z_{ik} = \begin{cases} 1, & \text{if component } x_i \text{ is assigned to ordered position } k, \\ 0, & \text{otherwise.} \end{cases}$$

$$\mathcal{O}(\mathbf{x}) = \left\{ \mathbf{z} \in \{0, 1\}^{n \times n} : \begin{aligned} & \sum_{i=1}^n z_{ik} = 1, \quad k \in N, \\ & \sum_{k=1}^n z_{ik} = 1, \quad i \in N, \\ & \sum_{i=1}^n z_{ik} x_i \leq \sum_{i=1}^n z_{i,k+1} x_i, \quad k < n \end{aligned} \right\}.$$

Each feasible $\mathbf{z} \in \mathcal{O}(\mathbf{x})$ assigns every component to one sorted position.

Integer Quadratic Formulation

Once the ordering is encoded by z , we have

$$x_{(k)} = \sum_{i=1}^n z_{ik} x_i.$$

Therefore,

$$Q(\mathbf{x}, M) = \min_{z \in \mathcal{O}(\mathbf{x})} \sum_{i=1}^n \sum_{k=1}^n \sum_{j=1}^n \sum_{\ell=1}^n m_{k\ell} x_i x_j z_{ik} z_{j\ell}.$$

- This is an assignment-based formulation.
- It is quadratic because of the products $z_{ik} z_{j\ell}$.
- Ties may produce multiple feasible orderings, but the value of Q is unchanged.

Linearization via Product Variables

Introduce

$$y_{ijk\ell} = z_{ik} z_{j\ell}, \quad i, j, k, \ell \in N.$$

$$\begin{aligned} Q(\mathbf{x}, M) = \min & \sum_{i,k=1}^n \sum_{j,\ell=1}^n m_{k\ell} x_i x_j y_{ijk\ell} \\ \text{s.t. } & z \in \mathcal{O}(\mathbf{x}), \\ & y_{ijk\ell} \leq z_{ik}, & i, j, k, \ell \in N, \\ & y_{ijk\ell} \leq z_{j\ell}, & i, j, k, \ell \in N, \\ & y_{ijk\ell} \geq z_{ik} + z_{j\ell} - 1, & i, j, k, \ell \in N, \\ & y_{ijk\ell} \geq 0, & i, j, k, \ell \in N. \end{aligned}$$

Size and Computational Implications

Model	Binary vars.	Continuous vars.	Constraints
Assignment ordering	$\mathcal{O}(n^2)$	$\circ$	$\mathcal{O}(n)$
Quadratic assignment	$\mathcal{O}(n^2)$	$\circ$	$\mathcal{O}(n)$
Linearized quadratic	$\mathcal{O}(n^2)$	$\mathcal{O}(n^4)$	$\mathcal{O}(n^4)$

- The formulation is exact and general.
- The main difficulty is the number of product variables $y_{ijk\ell}$.
- The structure is closely related to the Quadratic Assignment Problem.
- More specialized linearizations may be used to reduce computational burden.

Including Linear Ordered Measures

Quadratic ordered measures can be extended to include a linear ordered component:

$$\mathcal{Q}(\mathbf{x}, (\lambda|M)) = \sum_{k=1}^n \lambda_k x_{(k)} + \sum_{i=1}^n \sum_{j=1}^n m_{ij} x_{(i)} x_{(j)}.$$

- If $M = 0$, then

$$\mathcal{Q}(\mathbf{x}, (\lambda|M)) = \ell(\mathbf{x}, \lambda).$$

- If $\lambda = 0$, then

$$\mathcal{Q}(\mathbf{x}, (\boldsymbol{\lambda}|M)) = \mathcal{Q}(\mathbf{x}, M).$$

The combined linear-quadratic ordered measure can be computed as

$$\min_{\mathbf{z} \in \mathcal{O}(\mathbf{x})} \sum_{i=1}^n \sum_{k=1}^n \lambda_k x_i z_{ik} + \sum_{i,k=1}^n \sum_{\ell=1}^n m_{k\ell} x_i x_j z_{ik} z_{j\ell}.$$

Nested Ordered Measures

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Nested Ordered Measures

Many important dispersion and shape measures are not purely quadratic ordered measures.

They have a **nested structure**:

$$\text{complex measure} = \rho \left(\mathbf{x}, \underbrace{\ell(\mathbf{x}, \lambda)}_{\text{ordered location}} \right).$$

- Variance nests the mean.
- Mean Absolute Deviation nests the mean.
- Shape measures such as skewness and kurtosis also rely on a nested location.
- More generally, the nested location can be any $\mathcal{L}$ -measure.

General Nested Structure

Given $\mathbf{x} \in \mathbb{R}_+^n$ and $\lambda \in \mathbb{R}^n$, define

$$\mathfrak{b}(\mathbf{x}, \rho, \lambda) = \rho \left(\mathbf{x}, \sum_{k=1}^n \lambda_k x_{(k)} \right).$$

Equivalently, introducing an auxiliary variable θ ,

$$\begin{aligned} \mathfrak{b}(\mathbf{x}, \rho) = \min \quad & \rho(\mathbf{x}, \theta) \\ \text{s.t.} \quad & \theta = \ell(\mathbf{x}, \lambda). \end{aligned}$$

- θ represents the nested ordered location.
- The loss ρ measures deviations, dispersion, or shape around θ .
- The difficulty is the constraint

$$\theta = \ell(\mathbf{x}, \lambda).$$

Bilevel Interpretation

The nested measure can be viewed as a bilevel optimization problem.

Upper level

$$\min \rho(\mathbf{x}, \theta)$$

Lower level

$$\theta = \ell(\mathbf{x}, \lambda)$$

- The upper level evaluates the loss around θ .
- The lower level computes the ordered measure.
- Since the lower level has linear optimization formulations, it can be replaced by primal-dual optimality conditions.

Strong-Duality Reformulation

$$\ell(\mathbf{x}, \lambda) = \sum_{k=1}^n \Delta_k S_k(\mathbf{x}), \quad \Delta_k = \lambda_k - \lambda_{k-1}.$$

To impose

$$\theta = \ell(\mathbf{x}, \boldsymbol{\lambda}),$$

we impose two equivalent representations of the same lower-level optimal value.

- one using primal variables for some terms and dual variables for the others;
- the other using the opposite representation;
- equality of both values enforces optimality by strong duality.

Nested Model Based on k -sums

$$b_1(\mathbf{x}, \rho) = \min_{\theta} \rho(\mathbf{x}, \theta)$$

$$\text{s.t. } \theta = \sum_{k \in \Delta^+} |\Delta_k| \left((n-k+1)t_k + \sum_{i=1}^n v_{ik} \right) - \sum_{k \in \Delta^-} |\Delta_k| \sum_{i=1}^n d_{ik} x_i,$$

$$\theta = - \sum_{k \in \Lambda^-} |\Delta_k| \left((n-k+1)t_k + \sum_{i=1}^n v_{ik} \right) + \sum_{k \in \Lambda^+} |\Delta_k| \sum_{i=1}^n d_{ik} x_i,$$

$$t_k + v_{ik} \geq x_i, \quad i, k \in N,$$

$$\sum_{i=1}^n d_{ik} = n - k + 1, \quad k \in N,$$

$$t_k \in \mathbb{R}, \quad v_{ik} \geq 0, \quad i, k \in N,$$

$$0 \leq d_{ij} \leq 1, \quad i, j \in N,$$

$$\theta \geq 0.$$

How to Read the k -sum Nested Model

- The variable θ stores the value of the nested ordered measure:

$$\theta = \ell(\mathbf{x}, \boldsymbol{\lambda}).$$

- The variables (t_k, v_{ik}) represent the dual formulation of the k -sum operator:

$$S_k(\mathbf{x}) = \min \left\{ (n - k + 1)t_k + \sum_i v_{ik} : t_k + v_{ik} \geq x_i \right\}.$$

- The variables d_{ik} represent the primal selection formulation:

$$S_k(\mathbf{x}) = \max \left\{ \sum_i d_{ik} x_i : \sum_i d_{ik} = n - k + 1, 0 \leq d_{ik} \leq 1 \right\}.$$

- The two equations defining θ enforce primal-dual consistency.

Key point: The lower-level ordered measure disappears and is replaced by linear constraints.

Nested Model Based on Quantile Loss

$$\mathfrak{b}_2(\mathbf{x}, \rho) = \min_{\theta} \rho(\mathbf{x}, \theta)$$

$$\text{s.t. } \theta = \sum_{k \in \Delta^+} \sum_{i \in N} |\Delta_k| \left(\frac{k}{n} u_{ik} + \left(1 - \frac{k}{n} \right) v_{ik} \right) + \sum_{k \in \Delta^-} \sum_{i \in N} |\Delta_k| \alpha_{ik} x_i + c(\mathbf{x}),$$

$$\Theta = - \sum_{k \in \Delta^-} \sum_{i \in N} |\Delta_k| \left(\frac{k}{n} u_{ik} + \left(1 - \frac{k}{n} \right) v_{ik} \right) - \sum_{k \in \Delta^+} \sum_{i \in N} |\Delta_k| \alpha_{ik} x_i + c(\mathbf{x}),$$

$$u_{ik} - v_{ik} + \gamma_k = x_i, \quad i, k \in N,$$

$$\sum_{i \in N} \alpha_{ik} = 0, \quad k \in N,$$

$$\gamma_k \in \mathbb{R}, \quad u_{ik}, v_{ik} \geq 0, \quad i, k \in N,$$

$$-\frac{k}{n} \leq \alpha_{ik} \leq 1 - \frac{k}{n}, \quad i, k \in N,$$

$$\theta \geq 0.$$

What Remains Nonlinear?

Both nested formulations have **linear constraints**.

$$b_i(\mathbf{x}, \rho) = \min_{\theta} \rho(\mathbf{x}, \theta) \quad \text{s.t. linear constraints,} \quad i = 1, 2.$$

- If ρ is linear, the whole model is linear.
- If ρ is absolute deviation, the model can be linearized.
- If ρ is quadratic deviation, the model becomes convex quadratic when appropriate.
- If ρ represents skewness or kurtosis, higher-order nonlinearities remain in the objective.

Example: Mean Absolute Deviation

$$\rho(\mathbf{x}, \theta) = \sum_{i=1}^n |x_i - \theta|.$$

$$\text{MeanAD}_{\lambda}(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n \left| x_i - \sum_{k=1}^n \lambda_k x_{(k)} \right|.$$

- If $\lambda = (1/n, \dots, 1/n)$, the nested location is the mean.
- Other choices of λ give deviations from medians, quantiles, trimmed means, or OWA-type locations.

Linearizing Absolute Deviations

Introduce variables $r_i^+, r_i^- \geq 0$ such that

$$x_i - \theta = r_i^+ - r_i^-.$$

Then

$$\begin{aligned} |x_i - \theta| = \min \quad & r_i^+ + r_i^- \\ \text{s.t.} \quad & r_i^+ - r_i^- = x_i - \theta, \\ & r_i^+, r_i^- \geq 0. \end{aligned}$$

- At optimality, at most one of r_i^+ and r_i^- is positive.
- No complementarity constraint is needed.
- The minimization of $r_i^+ + r_i^-$ enforces the absolute value.

MAD with Nested Ordered Location

Combining the absolute-deviation linearization with either nested formulation gives

$$\begin{aligned} \text{MeanAD}_\lambda(\mathbf{x}) = \min \quad & \frac{1}{n} \sum_{i=1}^n (r_i^+ + r_i^-) \\ \text{s.t.} \quad & r_i^+ - r_i^- = x_i - \theta, & i \in N, \\ & r_i^+, r_i^- \geq 0, & i \in N, \\ & \theta = \ell(\mathbf{x}, \lambda). \end{aligned}$$

The last constraint can be replaced by:

- the k -sum strong-duality formulation; or
- the quantile-loss strong-duality formulation.

Special Nested Locations

$\theta = \ell(\mathbf{x}, \lambda)$

• Mean

$$\theta = \frac{1}{n} \sum_{i=1}^n x_i.$$

• Maximum

$$\theta = \max_i x_i.$$

• Minimum

$$\theta = \min_i x_i.$$

• Median

$$\theta = \text{median}(\mathbf{x}).$$

• Quantile

$$\theta = \text{quantile}_\tau(\mathbf{x}).$$

• Trimmed mean

$$\theta = \sum_k \lambda_k x_{(k)}.$$

Nested vs Quadratic Ordered Measures

	Quadratic ordered	Nested ordered
General form	$\sum_{i,j} m_{ij} x_{(i)} x_{(j)}$	$\rho(\mathbf{x}, \ell(\mathbf{x}, \lambda))$
Ordering mechanism	assignment of ordered positions	linear ordered measure inside loss
Typical examples	variance, trimmed variance, interactions	MAD, variance, skewness, kurtosis
Main difficulty	many pairwise products	nonlinearity of ρ
Modeling advantage	very expressive	often much more compact

Computational Insights

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Computational Experiments

We evaluate the optimization-based formulations for computing:

- linear ordered measures: ℓ_1, ℓ_2, ℓ_3 ;
- quadratic ordered measures: $\mathcal{Q}$;
- nested ordered measures: $\mathfrak{b}_1, \mathfrak{b}_2$.

$$n \in \{10, 20, 30, 40, 50, 100, 500, 1000\}, \quad x_i \sim U[0, 100].$$

- 20 independent instances per size.
- Python 3.10 and Gurobi 9.5.
- Time limit: 3600 seconds.
- Metrics: average CPU time and accuracy against standard Python routines.

Performance Metrics

Two indicators are used.

Time

Average CPU time required to compute the measure over the 20 instances.

Accuracy

Number of instances matching the Python reference value.

An instance is counted as accurate if

$$f_{python} - f_m \leq 10^{-5}.$$

Linear Ordered Measures

Measure	Model	Time								Accuracy							
		10	20	30	40	50	100	500	1000	10	20	30	40	50	100	500	1000
mean	ℓ_1	0.0012	0.0055	0.0173	0.0437	0.1533	52.5193	3600.0000	3600.0000	20	20	20	20	20	20	20	20
	ℓ_2	0.0001	0.0001	0.0002	0.0002	0.0002	0.0007	0.0051	0.0081	20	20	20	20	20	20	20	20
	ℓ_3	0.0001	0.0001	0.0001	0.0001	0.0001	0.0002	0.0029	0.0005	20	20	20	20	20	20	20	20
trim-mean	ℓ_1	0.0011	0.0055	0.0173	0.0435	0.1236	83.1352	3600.0000	3600.0000	20	20	20	20	20	20	0	0
	ℓ_2	0.0001	0.0001	0.0002	0.0002	0.0003	0.0007	0.0083	0.0118	20	20	20	20	20	20	20	20
	ℓ_3	0.0001	0.0001	0.0002	0.0002	0.0003	0.0009	0.0103	0.0106	20	20	20	20	20	20	20	20
win-mean	ℓ_1	0.0012	0.0055	0.0172	0.0437	0.1450	55.1646	3600.0000	3600.0000	20	20	20	20	20	20	0	0
	ℓ_2	0.0002	0.0002	0.0003	0.0003	0.0004	0.0014	0.0097	0.0195	20	20	20	20	20	20	20	20
	ℓ_3	0.0002	0.0002	0.0003	0.0004	0.0005	0.0015	0.0155	0.0281	20	20	20	20	20	20	20	20
range	ℓ_1	0.0011	0.0055	0.0173	0.0436	0.1528	52.6488	3600.0000	3600.0000	20	20	20	20	20	20	0	0
	ℓ_2	0.0002	0.0002	0.0002	0.0003	0.0003	0.0010	0.0064	0.0090	20	20	20	20	20	20	20	20
	ℓ_3	0.0002	0.0002	0.0003	0.0003	0.0004	0.0014	0.0094	0.0162	20	20	20	20	20	20	20	20
mid-mean	ℓ_1	0.0011	0.0055	0.0174	0.0437	0.1464	55.8178	3600.0000	3600.0000	20	20	20	20	20	20	0	0
	ℓ_2	0.0002	0.0002	0.0002	0.0003	0.0003	0.0007	0.0064	0.0154	20	20	20	20	20	20	20	20
	ℓ_3	0.0002	0.0002	0.0002	0.0003	0.0004	0.0010	0.0088	0.0122	20	20	20	20	20	20	20	20
median	ℓ_1	0.0011	0.0054	0.0175	0.0434	0.1458	3.2773	3600.0000	3600.0000	20	20	20	20	20	20	0	0
	ℓ_2	0.0002	0.0001	0.0002	0.0003	0.0003	0.0012	0.0088	0.0148	20	20	20	20	20	20	20	20
	ℓ_3	0.0001	0.0002	0.0002	0.0003	0.0003	0.0013	0.0142	0.0220	20	20	20	20	20	20	20	20
1quartile	ℓ_1	0.0011	0.0055	0.0172	0.0438	0.1665	107.7195	3600.0000	3600.0000	20	20	20	20	20	20	0	0
	ℓ_2	0.0002	0.0002	0.0002	0.0002	0.0003	0.0009	0.0086	0.0141	20	20	20	20	20	20	20	20
	ℓ_3	0.0001	0.0001	0.0002	0.0002	0.0003	0.0010	0.0119	0.0157	20	20	20	20	20	20	20	20
3quartile	ℓ_1	0.0012	0.0055	0.0172	0.0436	0.1350	51.9655	3600.0000	3600.0000	20	20	20	20	20	20	0	0
	ℓ_2	0.0002	0.0001	0.0002	0.0002	0.0003	0.0009	0.0070	0.0106	20	20	20	20	20	20	20	20
	ℓ_3	0.0001	0.0002	0.0002	0.0003	0.0003	0.0012	0.0122	0.0158	20	20	20	20	20	20	20	20
iqr	ℓ_1	0.0012	0.0055	0.0170	0.0433	0.1405	58.9756	3600.0000	3600.0000	20	20	20	20	20	20	0	0
	ℓ_2	0.0002	0.0003	0.0003	0.0004	0.0005	0.0013	0.0129	0.0325	20	20	20	20	20	20	20	20
	ℓ_3	0.0002	0.0002	0.0003	0.0005	0.0006	0.0018	0.0178	0.0426	20	20	20	20	20	20	20	20
madev-median	ℓ_1	0.0011	0.0054	0.0172	0.0435	0.1446	52.9097	3600.0000	3600.0000	20	20	20	20	20	20	0	0
	ℓ_2	0.0001	0.0002	0.0002	0.0003	0.0003	0.0008	0.0085	0.0143	20	20	20	20	20	20	20	20
	ℓ_3	0.0001	0.0002	0.0002	0.0003	0.0003	0.0009	0.0112	0.0151	20	20	20	20	20	20	20	20

Linear Ordered Measures: Main Findings

- ℓ_1 performs well for small values of n .
- However, ℓ_1 scales poorly and frequently reaches the time limit for large instances.
- ℓ_2 and ℓ_3 are much faster and remain accurate for all tested sizes.
- For $n = 1000$, ℓ_2 and ℓ_3 solve most measures in milliseconds or a few hundredths of a second.
- The only relatively heavier linear measure is gini-difs, but it is still solved efficiently by ℓ_2 and ℓ_3 .

Quadratic Ordered Measures

Measure	Model	Time					Accuracy				
		10	20	30	40	50	10	20	30	40	50
variance	Q	9.1381	13.6615	128.1267	902.3868	3292.905	20	20	20	20	20
msdev-median	Q	2.8683	13.9078	125.0910	793.5474	2187.096	20	20	20	20	20
msdev-max	Q	2.1225	12.4740	118.8592	746.7027	2225.038	20	20	20	20	20
msdev-min	Q	1.7471	12.2372	118.0600	750.9281	2199.251	20	20	20	20	20
trim-variance	Q	11.0785	13.0243	119.2869	844.5739	2952.002	20	20	20	20	20
win-variance	Q	10.5823	12.8667	120.0339	855.3233	3224.855	20	20	20	20	20

Quadratic Ordered Measures: Main Findings

- The $\mathcal{Q}$ formulation is exact and achieves full accuracy in all tested cases.
- Computational time increases rapidly with n .
- For variance, the average time grows from about 9 seconds for $n = 10$ to more than 3000 seconds for $n = 50$.
- Similar behavior appears for trimmed and winsorized variances.
- This reflects the high cost of modeling all ordered pairwise interactions.

Nested Ordered Measures

Measure	Model	Time								Accuracy							
		10	20	30	40	50	100	500	1000	10	20	30	40	50	100	500	1000
variance	b ₁	0.0003	0.0005	0.0009	0.0014	0.0021	0.0098	0.3784	1.8353	20	20	20	20	20	20	20	20
	b ₂	0.0002	0.0004	0.0009	0.0016	0.0024	0.0107	0.3871	2.1006	20	20	20	20	20	20	20	20
msdev-median	b ₁	0.0004	0.0008	0.0013	0.0023	0.0034	0.0131	0.3337	1.7240	20	20	20	20	20	20	20	20
	b ₂	0.0004	0.0009	0.0016	0.0029	0.0045	0.0166	0.3927	2.0073	20	20	20	20	20	20	20	20
msdev-max	b ₁	0.0003	0.0006	0.0010	0.0017	0.0025	0.0107	0.3611	1.6875	20	20	20	20	20	20	20	20
	b ₂	0.0003	0.0007	0.0013	0.0023	0.0034	0.0132	0.4167	1.9957	20	20	20	20	20	20	20	20
msdev-min	b ₁	0.0004	0.0007	0.0013	0.0023	0.0034	0.0136	0.3471	1.8349	20	20	20	20	20	19	14	10
	b ₂	0.0003	0.0007	0.0012	0.0021	0.0034	0.0130	0.4140	1.9671	20	20	20	20	20	20	20	20
madev-mean	b ₁	0.0006	0.0014	0.0029	0.0054	0.0129	0.1243	0.5407	2.6555	20	20	20	20	20	20	20	20
	b ₂	0.0003	0.0006	0.0011	0.0020	0.0032	0.0152	0.5710	2.7334	20	20	20	20	20	20	20	20
madev-median	b ₁	0.0011	0.0019	0.0031	0.0047	0.0066	0.0270	0.6400	3.0635	20	20	20	20	20	20	20	20
	b ₂	0.0012	0.0021	0.0035	0.0054	0.0075	0.0279	0.8018	3.5806	20	20	20	20	20	20	20	20
madev-max	b ₁	0.0008	0.0012	0.0016	0.0025	0.0036	0.0167	6.4872	51.1322	20	20	20	20	20	20	20	20
	b ₂	0.0008	0.0017	0.0031	0.0067	0.0104	0.0308	0.7752	3.9238	20	20	20	20	20	20	20	20
madev-min	b ₁	0.0013	0.0034	0.0074	0.0093	0.0133	0.0439	0.7684	3.2349	20	20	20	20	20	20	20	20
	b ₂	0.0005	0.0009	0.0015	0.0028	0.0041	0.0180	0.6795	3.2782	20	20	20	20	20	20	20	20
skewness	b ₁	0.0018	0.0030	0.0054	0.0099	0.0204	0.1950	0.9368	52.8096	20	20	20	20	20	20	20	20
	b ₂	0.0014	0.0019	0.0031	0.0050	0.0073	0.0286	10.4145	5.0563	20	20	20	20	20	20	20	20
kurtosis	b ₁	0.0016	0.0029	0.0053	0.0095	0.0205	0.2081	0.9167	4.6692	20	20	20	20	20	20	20	20
	b ₂	0.0012	0.0019	0.0031	0.0051	0.0073	0.0302	1.0423	5.0956	20	20	20	20	20	20	20	20

Nested Ordered Measures: Main Findings

- Both b_1 and b_2 are extremely fast compared with the full quadratic model.
- They maintain full accuracy in almost all tested cases.
- The only notable exception is $msdev-min$, where b_2 remains fully accurate while b_1 loses accuracy for large n .
- For several large instances, b_2 is faster than b_1 , especially for $madev-max$ and skewness.
- Both formulations scale up to $n = 1000$.

Quadratic vs Nested Computation

The most striking comparison appears for the variance.

Quadratic model Q

For $n = 50$:

time > 3000 seconds.

Nested models b_1, b_2

For $n = 1000$:

time ≈ 2 seconds.

Main message

Measures that appear to require quadratic ordered formulations can often be computed much more efficiently through nested ordered representations.

Scalability Patterns

